

Real-Time FDAS Alarm Notification System via SMS

Camera-based OCR detection • automatic SMS dispatch • permanent event logging

Proposed Solution

The proposed solution observes the panel visually, in the same manner as a human operator, using a fixed-position camera. The system extracts the alarm's unique device address code through image processing and OCR, cross-references it against a pre-configured location database, and automatically dispatches an SMS notification to the responsible personnel for that location. Every detected event is logged permanently, independent of SMS delivery outcome. The system is designed as a supplementary notification layer and does not modify, interface with, or replace the certified F DAS panel, hooters, or sirens.

COMPONENT	Specification / Role
FDAS Panel	Existing certified fire alarm panel (unmodified). LCD display and LED indicators serve as the sole data source.
Camera	Raspberry Pi HQ Camera (Sony IMX477, 12.3MP) with 8mm CS-mount manual-focus lens; fixed mounting bracket; camera adapter cable.
Optical accessory	Anti-glare hood / polarizing filter to reduce reflection from the panel's glossy display surface.
Microprocessor	Raspberry Pi 5, UPS-backed power supply, running the detection and notification pipeline locally at the installation site
Communication module	GSM module (e.g. SIM7600) or cloud SMS gateway API for alert dispatch.

Software Pipeline — Detailed Design

01 Frame Acquisition

Video frames are continuously sampled from the camera feed at approximately 1 frame per second. Frames that are fully black or overexposed are discarded prior to further processing.

02 Primary Trigger — LED State Detection

A fixed pixel region corresponding to the FIRE LED is sampled every frame and classified by color (HSV thresholding). Downstream processing is invoked only on an OFF-to-RED state transition, minimizing continuous computational load.

03 Region of Interest (ROI) Extraction

A statically defined bounding box, established during calibration, is used to crop the display region containing the device address code, avoiding full-frame processing.

04 Image Preprocessing

The cropped region undergoes grayscale conversion, contrast enhancement, and adaptive binarization to improve text legibility prior to OCR.

05 Optical Character Recognition

The preprocessed image is passed to an OCR engine (PaddleOCR, with Tesseract as fallback) to extract the device address code as text, along with a per-character confidence score.

06 Validation

Extracted text is validated against (a) an expected format pattern via regular expression, and (b) a master list of known device addresses. Non-conforming results are discarded and re-attempted on the subsequent frame.

07 Debounce / Confirmation

A validated reading must recur consistently across 2–3 consecutive frames before being accepted, mitigating false triggers from transient glare or reflection.

08 Event Deduplication

An in-memory active-event registry prevents repeated notifications for an alarm that remains displayed; only new or state-changed events proceed to notification.

09 Location Resolution

The confirmed device code is cross-referenced against a commissioning-time mapping table to resolve physical location, zone, device type, and the applicable contact group.

10 Event Logging

The structured event record is committed to a persistent local database prior to any notification attempt, ensuring an audit trail independent of notification outcome.

11 Notification Dispatch

An SMS is dispatched to the resolved contact group via GSM module or SMS gateway API. Failed transmissions are retried with backoff and escalated to a secondary channel if unsuccessful.

12 Event Resolution Tracking

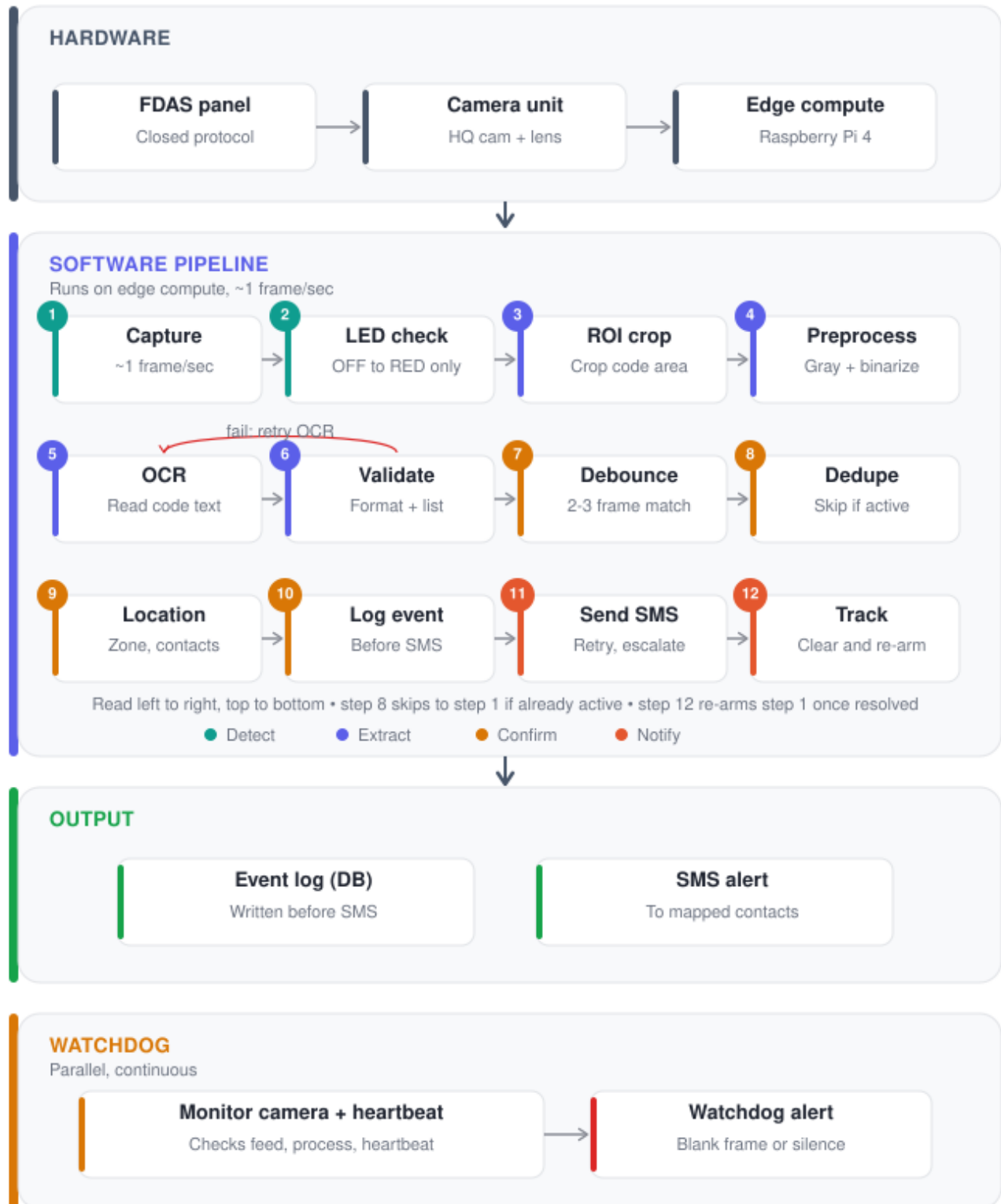
The system continues monitoring the active event; upon its disappearance from the panel display, the event is marked resolved and the deduplication state is reset for that device code.

13 System Health Monitoring

An independent, continuously running process monitors the health of the acquisition and processing pipeline — verifying frame freshness, expected brightness range, and periodic heartbeat signals. A distinct system-fault alert is raised should any check fail, ensuring that a failure of the monitoring apparatus itself is not mistaken for normal operation.

FDAS alarm notification system

Architecture overview — hardware, software pipeline, output, and watchdog



Slate = hardware • colored badges = pipeline phase • green = output • amber/red = watchdog